

TRAINING-FREE TEMPORAL VISUAL CACHING IN OPENVLA-OFT: A CLOSED-LOOP EVALUATION

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INTRODUCTION

- VLA policies repeatedly process similar camera views.
- Reuse may save work, but stale visual states can change actions.
- Goal: lower query time without reducing task success.

METHODS



Complete prefix
reuse both
cameras

Scene camera
refresh wrist
camera

Internal K/V
reuse layers
+ regions

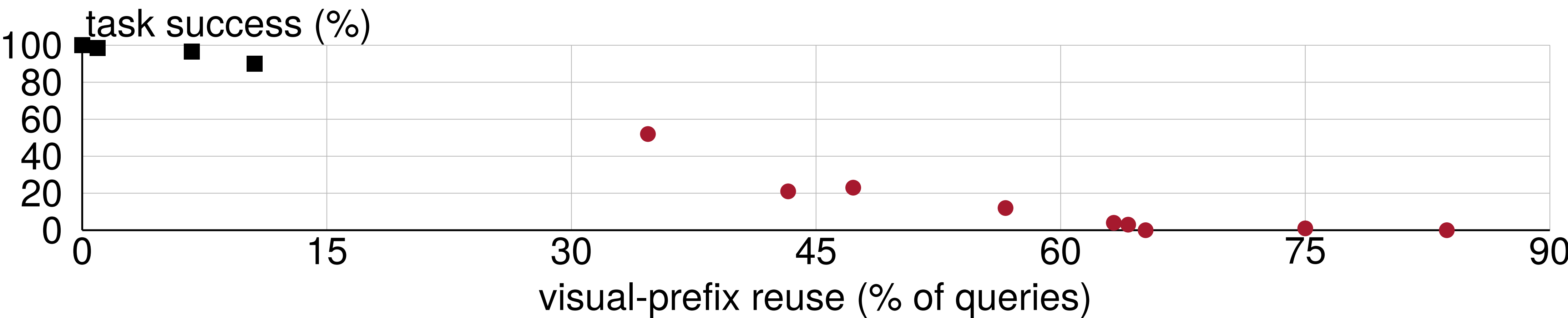
Compared with dense inference using
task success, reuse rate, and complete query time.

OpenVLA-OFT | LIBERO-Spatial | TITAN RTX

RESULTS

COMPLETE VISUAL-PREFIX REUSE

• permissive reuse ■ dense / conservative



Dense: **100/100**. Reusing 34.69–83.68% of queries gave **0–52% success**.
Conservative reuse reached **69/70**, but reused only **0.95%** of queries.

SCENE-CAMERA REUSE

Task success

Dense **67/70**

Scene reuse **67/70**

Mean complete-query time

Dense

1,188.28 ms

Scene reuse

1,190.97 ms

Scene reused on 25.24% of queries.

Success: 67/70; query time: **+0.23%**.

FINE-GRAINED INTERNAL REUSE

Median complete-query time

Dense **100%**

Selective reuse **77.4%**

Closed-loop task success

Dense: 68/120

Selective reuse: **0/120**

22.60% lower median query time;
no successful cached episodes.

CONCLUSION

LIMITATIONS



The tested training-free reuse strategies
did not preserve task success while
reducing end-to-end query time.

- One model checkpoint
- One benchmark and GPU class
- Terminal task success only

REFERENCES

- [1] Kim et al., RSS 2025.
- [2] Liu et al., NeurIPS 2023.
- [3] Xu et al., NeurIPS 2025.

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CODE github.com/vwdr/SAVR

